



SCHOOL OF DATA SCIENCE

Data Analyst

Syllabus

UDACITY.COM

Overview

Clean up messy data, uncover patterns and insights, and communicate your findings. You'll start with an introduction to data analysis tools, including Jupyter Notebook, NumPy, pandas, and Matplotlib. Using these tools, you will ask questions about data and answer them through data collection, exploration, wrangling, and visualization. This intermediate-level program includes real-world projects where you will choose your own datasets, research questions, and analysis approach. As you progress through the program, each course will repeat the data analysis process while introducing more advanced techniques, such as applying data imputation to fill in missing data and applying appropriate encodings when developing data visualizations.

Nanodegree Program

Intermediate



43 hours



4.8 (1268 Reviews)

Prerequisites

Prior to enrolling, you should have the following knowledge:

File i/o

Inferential statistics

Basic Python

Elementary algebra

Basic SQL

Basic descriptive statistics

You will also need to be able to communicate fluently and professionally in written and spoken English.

Skills You'll Learn

Introduction to Data Analysis with Pandas and NumPy

Pandas | Exploratory data analysis | Basic data visualizations | Jupyter notebooks | Data storytelling | Data analysis process | NumPy | Data manipulation

Advanced Data Wrangling

Data cleaning | Data storage | Data Tidiness Assessment | Data gathering | Pandas | Data quality assessment | Data wrangling | File i/o

Data Visualization with Matplotlib and Seaborn

Latent variables | Data visualization design | Data fluency | Exploratory data analysis | Professional presentations | Data limitations and biases | Data storytelling | Jupyter notebooks | Quantitative data visualization

Courses

01. Welcome to the Data Analyst Nanodegree Program

15 minutes

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| L1 | An Introduction to Your Nanodegree Program | Welcome! We're so glad you're here. Join us in learning a bit more about what to expect and ways to succeed. |
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02. Introduction to Data Analysis with Pandas and NumPy

14 hours

Learn the data analysis process of questioning, wrangling, exploring, analyzing, and communicating data. You will work with data in Python using libraries like NumPy and pandas.

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| L1 | The Data Analysis Process | Learn about the data analysis process and the Python packages used in this course |
| L2 | Jupyter Notebooks | Jupyter Notebooks are a great tool for sharing insights and visualizations alongside your code. This lesson covers how to create them and utilize their various features. |
| L3 | Exploring and Inspecting Data | Use the pandas library to load data, view its properties, and start asking data analysis questions |
| L4 | Manipulating Data using Pandas and NumPy | Use the pandas library to perform data cleaning, filtering, and reshaping tasks. This includes troubleshooting issues with data as well as optimizing for memory usage and speed. |
| L5 | Communicating Results | Draw conclusions and communicate results to stakeholders by calculating statistics and creating basic data visualizations with the pandas library |
| L6 | Project:
Investigate a Dataset | Choose one of Udacity's curated datasets, perform an investigation, and share your findings. |

03. Advanced Data Wrangling

15 hours



Data wrangling is a set of processes for turning raw and messy data into a clean format to answer interesting questions from the data. In this course, you will learn the three phases of data wrangling: gathering, assessing, and cleaning data.

L1	Introduction to Data Wrangling	You will learn what data wrangling is and why it matters. And you will see a real-world example of data wrangling and some common misconceptions about data wrangling.
L2	Gathering Data	You will learn to implement data gathering methods to obtain and extract data from various sources and in several popular data formats.
L3	Assessing Data	You will learn to identify different data quality and structural issues and apply visual and programmatic assessments to catch them.
L4	Cleaning Data	You will learn to remediate the issues you identified in the assessment stage and test that your data cleaning is successful.
L5	Project: Real World Data Wrangling with Python	You will apply the skills you acquired in the course by gathering, assessing, and cleaning multiple real-world datasets of your choice.

04. Data Visualization with Matplotlib and Seaborn

15 hours

Learn to apply sound design and data visualization principles to the data analysis process. Learn how to use analysis and visualizations to tell a story with data.

L1	Data Visualization in Data Analysis	In this lesson, see the motivations for why data visualization is an important part of the data analysis process and where it fits in.
L2	Design of Visualizations	Learn about elements of visualization design, especially to avoid those elements that can cause a visualization to fail.
L3	Univariate Exploration of Data	In this lesson, you will see how you can use matplotlib and seaborn to produce informative visualizations of single variables.
L4	Bivariate Exploration of Data	In this lesson, build up from your understanding of individual variables and learn how to use matplotlib and seaborn to look at relationships between two variables.
L5	Multivariate Exploration of Data	In this lesson, see how you can use matplotlib and seaborn to visualize relationships and interactions between three or more variables.
L6	Explanatory Visualizations	Previous lessons covered how you could use visualizations to learn about your data. In this lesson, see how to polish up those plots to convey your findings to others!



L7

Project:

Communicate Data Findings

Choose a dataset, either your own or a Udacity-curated dataset, and perform an exploratory data analysis using Python. Then, create a presentation with explanatory plots that conveys your findings.

05. Congratulations!

10 minutes

Congratulations on finishing your program!

L1

Congratulations!

Congratulations on your graduation from this program! Please join us in celebrating your accomplishments.

Meet Your Instructors



Josh Magee

Senior Data Scientist

Josh is a Senior Data Scientist at Local Logic, where he models commercial real estate trends, acquisitions, and sustainable cities. He was formerly Assistant Professor of Data Analytics at Stonehill College, and was a postdoctoral researcher in nuclear physics at Lawrence Livermore National Laboratory.



Ria Cheruvu

AI Software Architect

Ria is an AI Software Architect and technical lead at Intel. She has a master's in data science from Harvard University, and is an accomplished industry speaker and instructor. She formerly served as Intel NEX's AI Ethics Lead Architect, leading trustworthy AI product creation, and as a Teaching Fellow for Harvard Data Science. Ria has multiple patents and publications on AI and ethics, and enjoys contributing to open-source communities to advance innovation.



Matt Maybeno

Principal Software Engineer

Matt is a Principal Software Engineer at SOCi. With a masters in Bioinformatics from SDSU, he utilizes his cross domain expertise to build solutions in NLP and predictive analytics.

Why Udacity



Demonstrate proficiency with practical projects

Projects are based on real-world scenarios and challenges, allowing you to apply the skills you learn to practical situations, while giving you real hands-on experience

- ✓ Gain proven experience
- ✓ Retain knowledge longer
- ✓ Apply new skills immediately



24/7 access to real human support

Reviewers provide timely and constructive feedback on your project submissions, highlighting areas of improvement and offering practical tips to enhance your work

- ✓ Get help from subject matter experts
- ✓ Gain valuable insights and improve your skills
- ✓ Learn industry best practices